

# LAB 3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

## Clinical Trial Subject Recruitment Engine

### 1. Scenario and Requirements Overview

The Clinical Trial Subject Recruitment Engine is used by research institutions to screen and manage candidates for clinical trials. It collects candidate information, checks eligibility, records informed consent and helps researchers keep track of trial visits and candidate status.

#### Core Functionality

1. Candidates can enter and submit their information for initial screening.
2. The system checks the candidate against the trial inclusion and exclusion criteria.
3. Candidates can provide and record informed consent.
4. Researchers can record and check trial visit compliance.
5. Researchers can view a candidate's eligibility, consent and visit status.

#### Technical Constraints

1. Candidate information must be protected from unauthorized access.
2. Electronic clinical trial records need a secure audit trail.
3. The recruitment database must follow 21 CFR Part 11 requirements.

#### Main Challenges

1. Security: protecting candidate and trial information.
2. Data integrity: keeping clinical records accurate and traceable.
3. Performance: keeping common screening and status operations reasonably quick.
4. Maintainability: keeping the main functions separated enough to make future changes easier.

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### 2. Architectural Style Analysis

The three architectural styles given in the lab manual are compared below for this scenario.

| Architecture Style         | Advantages                                                                                    | Disadvantages                                                                        | Suitability                                                           |
|----------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Layered Architecture       | Clear separation between interface, business logic and data. Easy to understand and maintain. | Requests can pass through several layers, which can add some overhead.               | High - fits the structured nature of the system.                      |
| Microservices Architecture | Functions can be split into separate services and scaled or changed independently.            | More complex to deploy and maintain. Service-to-service communication adds overhead. | Medium - useful for larger systems, but more than this project needs. |
| Client-Server Architecture | Centralized data and control make deployment and data management simple.                      | The central server can become a bottleneck and a single point of failure.            | Medium - workable, but less flexible for separating system functions. |

### 3. Architecture Selection and Component Identification

For this system, the selected architecture is Layered Architecture. The screening, consent and researcher functions are different parts of the system, but they all work with the same clinical trial data, which fits a layered structure well.

#### 3.1 Selected Architecture

1. Layered Architecture keeps the user-facing parts separate from the business rules and the database. This makes the system easier to follow and maintain.
2. It also gives a clear place to handle validation, access control and trial rules before the database is accessed.

#### 3.2 Components

| Component                            | Responsibility                                                                                 |
|--------------------------------------|------------------------------------------------------------------------------------------------|
| Candidate Interface Component        | Handles screening form entry and informed-consent actions for the trial candidate.             |
| Researcher Interface Component       | Lets the clinical researcher view candidate status and work with visit information.            |
| Screening & Eligibility Component    | Processes screening details and checks the candidate against inclusion and exclusion criteria. |
| Consent & Visit Management Component | Records informed consent and manages completed, missed and upcoming trial visits.              |
| Clinical Trial Database Component    | Stores candidate, screening, consent, visit and audit information.                             |

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### 4. UML Component Diagram

The component diagram below shows the five components, their main interfaces and the database dependencies in the selected layered architecture.

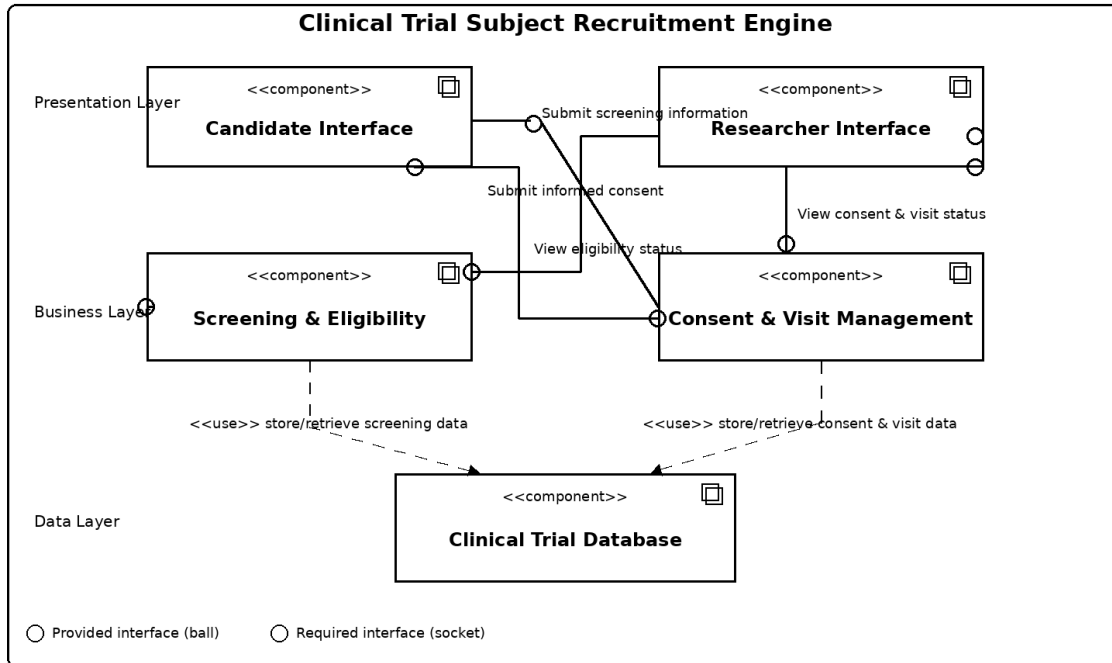


Figure 1. UML component diagram for the Clinical Trial Subject Recruitment Engine

#### 4.1 Interfaces

| Interface                                         | Interaction                                                      |
|---------------------------------------------------|------------------------------------------------------------------|
| Candidate Interface ↔ Screening & Eligibility     | Submit screening information and receive the eligibility result. |
| Candidate Interface ↔ Consent & Visit Management  | Submit informed consent information.                             |
| Researcher Interface ↔ Screening & Eligibility    | View candidate screening and eligibility status.                 |
| Researcher Interface ↔ Consent & Visit Management | View consent and visit compliance status.                        |

### 5. Architectural Justification

We chose Layered Architecture for the Clinical Trial Subject Recruitment Engine because it matches the way the system is divided into user interaction, processing and data storage. It gives the project a clear structure without adding the extra operational complexity that a microservices approach would bring.

1. Reason 1: The system has separate areas for candidate screening, eligibility, consent and visit tracking. Keeping these responsibilities in the business layer makes the design easier to understand and change later.
2. Reason 2: Candidate and trial information should not be accessed directly from the interface. A layered structure gives the application layer a place to perform validation and apply trial rules before data is stored or retrieved.
3. Security Advantage: Access checks can be handled before sensitive candidate records are returned. The database can also keep the audit information needed for traceability and the 21 CFR Part 11 requirement.
4. Performance Benefit: The system can handle common operations such as eligibility checks and candidate-status retrieval inside one application structure. This avoids the extra network communication and service-management overhead that would come with splitting the system into many small services.